

Copyright Notice

These slides are distributed under the Creative Commons License.

[DeepLearning.AI](#) makes these slides available for educational purposes. You may not use or distribute these slides for commercial purposes. You may make copies of these slides and use or distribute them for educational purposes as long as you cite [DeepLearning.AI](#) as the source of the slides.

For the rest of the details of the license, see <https://creativecommons.org/licenses/by-sa/2.0/legalcode>

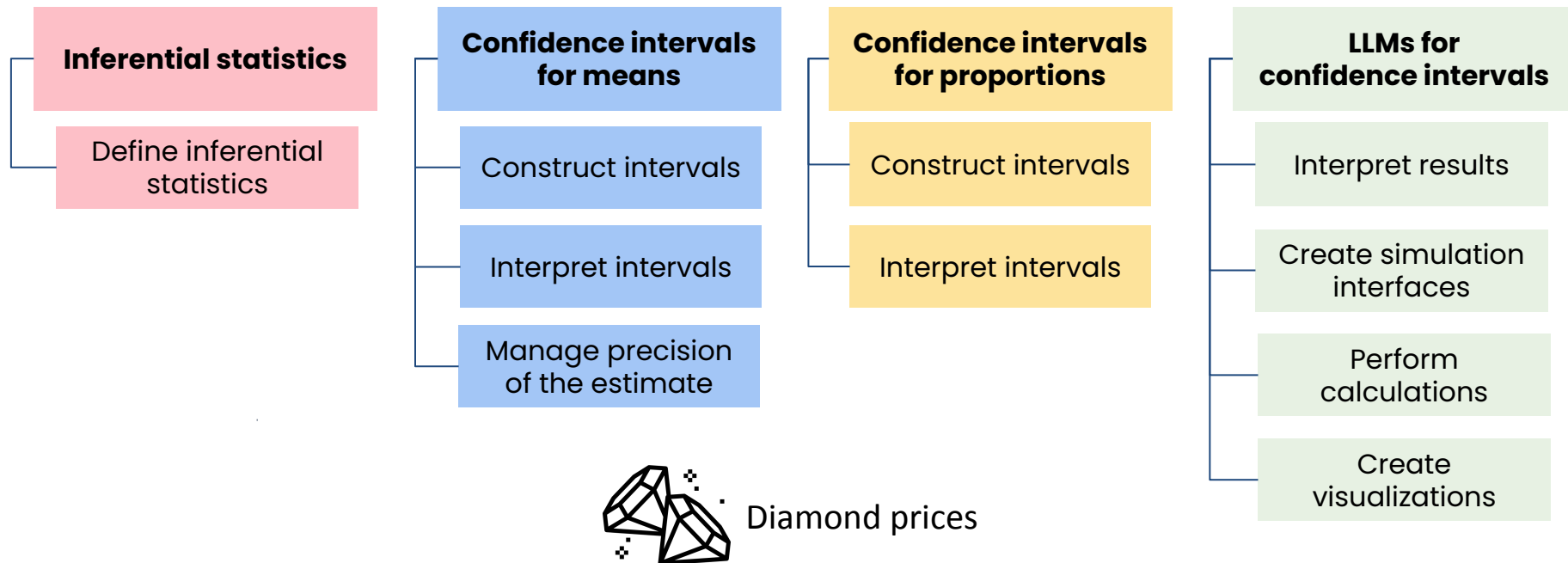
Applied Statistics for Data Analytics

Module 3: Confidence intervals

Confidence intervals

Module 3 introduction

Module 3 outline



Confidence intervals

Inferential statistics

Scenario

Figure out the rate of employee satisfaction:



You
Data Analyst



Satisfaction is likely high



100 employees ➡ **82.0%** satisfied ➤



500 employees ➡ **91.0%** satisfied



500 employees ➡ **91.0%** satisfied

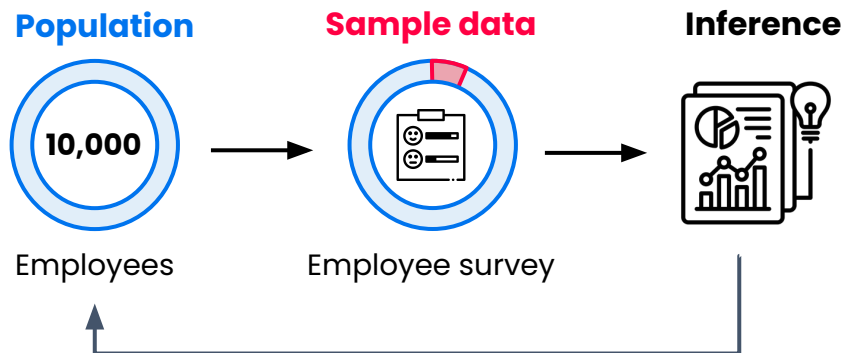


500 employees ➡ **92.1%** satisfied



500 employees ➡ **88.7%** satisfied

Inferential statistics



- Larger samples provide more reliable estimates than smaller samples



- Different samples show variability, even when drawn from the same population



- Allows you to quantify your level of confidence in your estimate

Descriptive statistics

State facts about sample

- "In a sample of 100 employees, 82 said they were satisfied."
- "A survey of 160 parents found that the parents with newborns got 6.1 hours of sleep while the parents with older children got 8.2 hours."

Describe the characteristics of a sample

Inferential statistics

Use sample data to draw conclusions about population

- Based on a sample of 100 employees, employee satisfaction among all 10,000 employees is likely between 88% and 91%.
- A survey of parents concluded that parents of older children sleep 2 hours more per night compared with parents of newborns.

Use characteristics of sample to make conclusions about population

Inferential statistics

- Use probability to draw conclusions about population based on sample statistics
 - Taking into account:



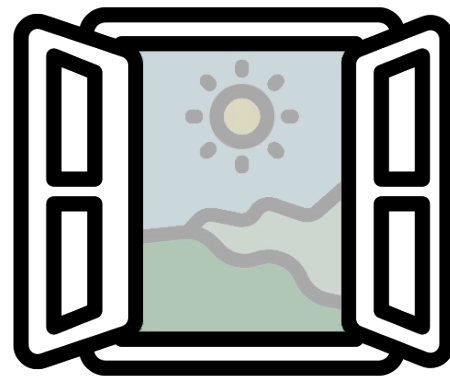
Size of sample



Variability of the sample

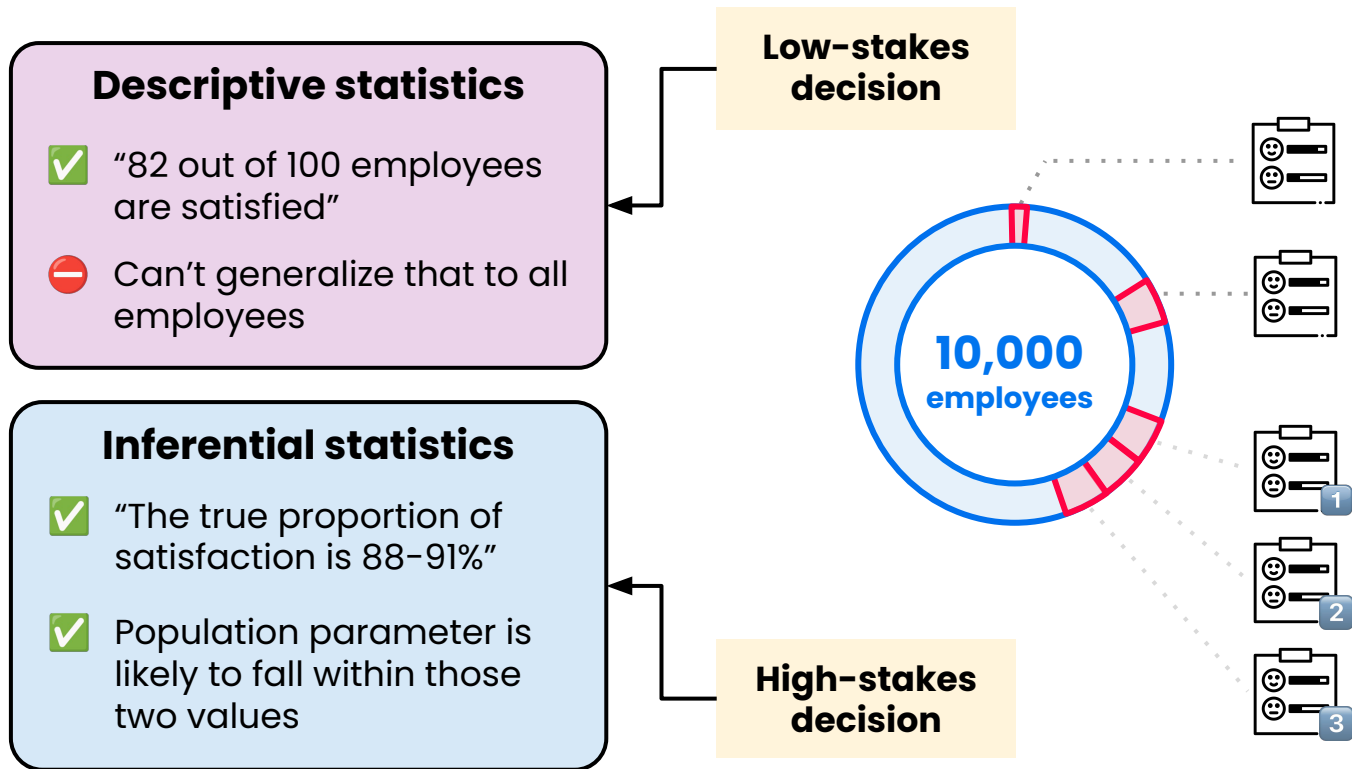


Other factors



Sample

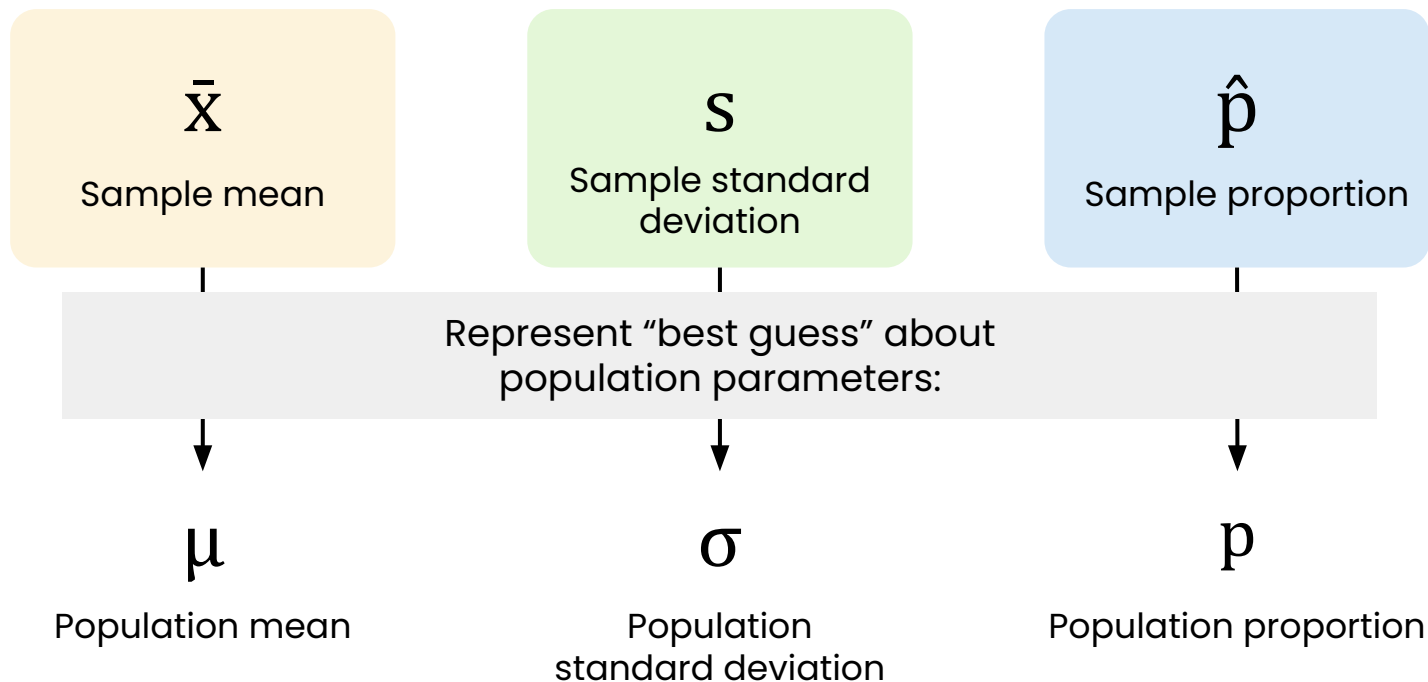
Inferential vs. descriptive statistics



Confidence intervals

Point & interval estimates

Point estimates



Point estimates

❌ **Don't** contain information about the confidence of estimate

Example:

Random sample of 25 movies from 2013

↳ $\bar{x}$ of durations = 121 minutes

How certain can you be that true population mean is **exactly** 121?



Interval estimates

✅ Contain information about how confident you can be

Example:

Arriving within 10 to 15 mins



Arriving in exactly 10 mins



Visualizing intervals

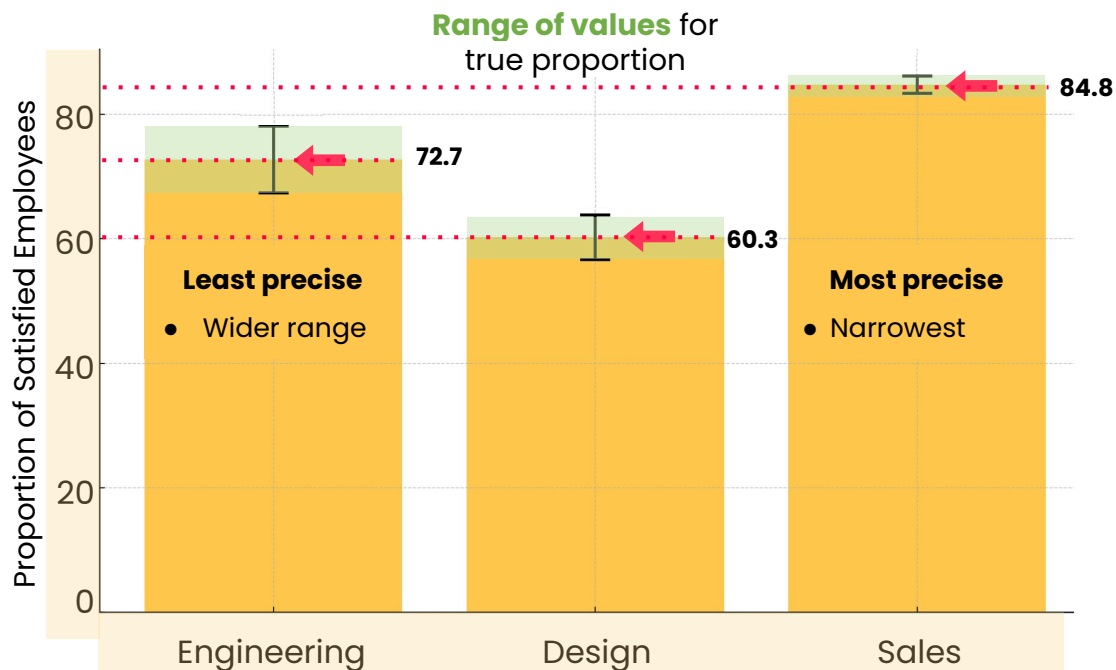
Error bars: visual representation of an **interval estimate**

- Inference about true population proportion based on sample



If repeated many times, expect true population value to fall within this range

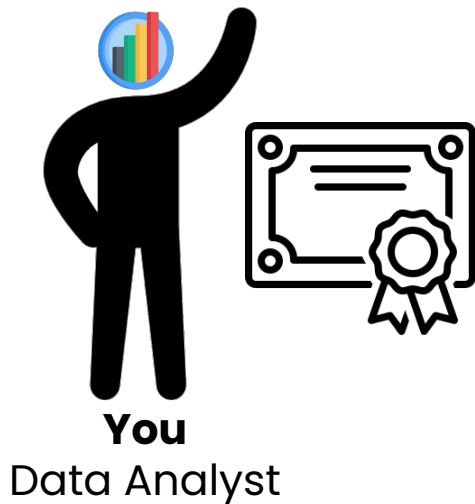
Employee Satisfaction by Department



Confidence intervals

Sampling distributions &
the central limit theorem

Scenario



Task: Estimate average score on a professional certification exam



Possible scores: 0 to 100

Sample 1



Ask **50 random people** their scores

$\bar{x}$

of scores

Sample 2



Ask **50 random people** their scores

$\bar{x}$

of scores

Same value?

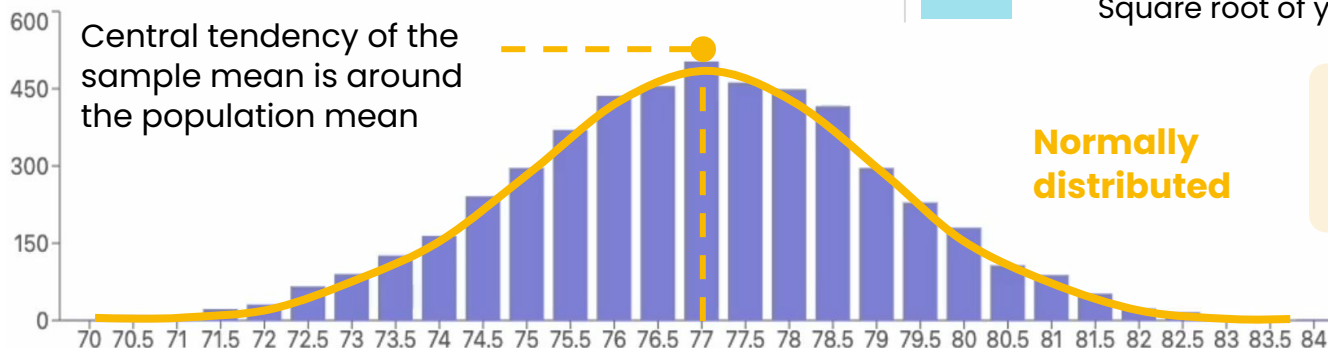
Central limit theorem



$n > 30$

If you take sufficiently large samples from any distribution and calculate their **means**:

- Sample means will be normally distributed
- Mean of distribution will equal μ



Increase your sample size



Estimate becomes more precise

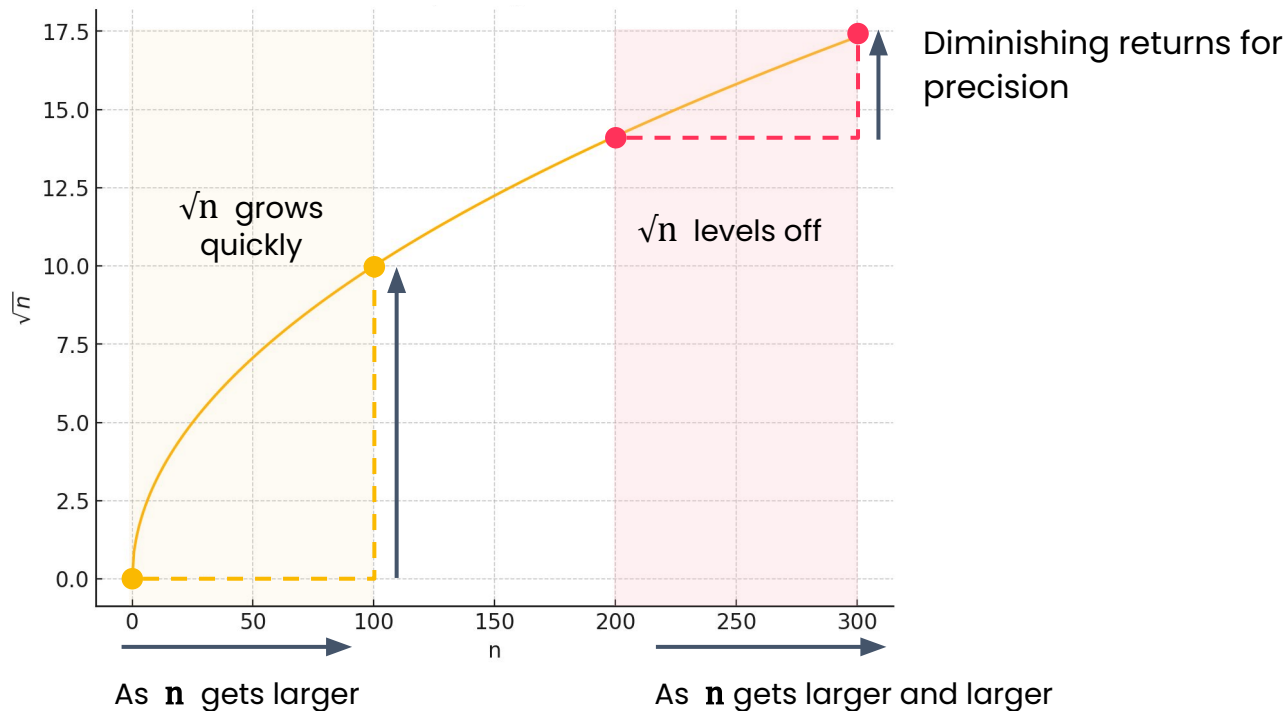
Standard deviation of your sample mean
(**standard error** of the mean)

$$SE = \frac{\text{True population standard deviation}}{\text{Square root of your sample size}} = \frac{\sigma}{\sqrt{n}}$$

As n gets larger, $\sqrt{n}$ gets larger, but at a **slower rate**

Sample size and precision

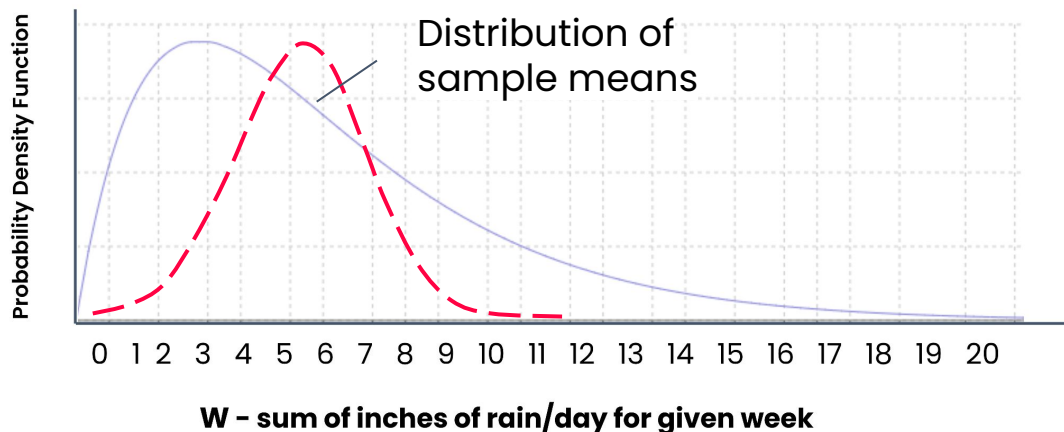
Large gains in precision from adding a few more values



Central limit theorem

Even if sample data is not normal:

- Sample means **will** be normal as long as n is sufficiently large



Central limit theorem applies to:

- Sample mean $\bar{X}$
- Sample proportion $\hat{p}$

In some cases:

- Sample variance s^2
- Sample standard deviation s

Confidence intervals

Demo: confidence intervals
in action

Confidence intervals

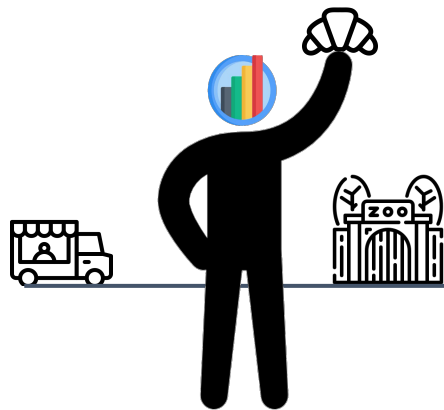


Confidence intervals

Scenario



Before 7am



You

Data Analyst

Task: Figure out how long it takes to deliver the pastries

Sample data:



Monitor the delivery truck for 30 days



Record time it takes to get from bakery to zoo

$\bar{x} = 43$ minutes

$s = 11$ minutes



Maybe due to chance
your deliveries were:



Unusually fast
Unusually slow

Confidence interval

$\bar{x} = 43$ minutes

$s = 11$ minutes



Best guess estimate for
mean delivery time (μ)

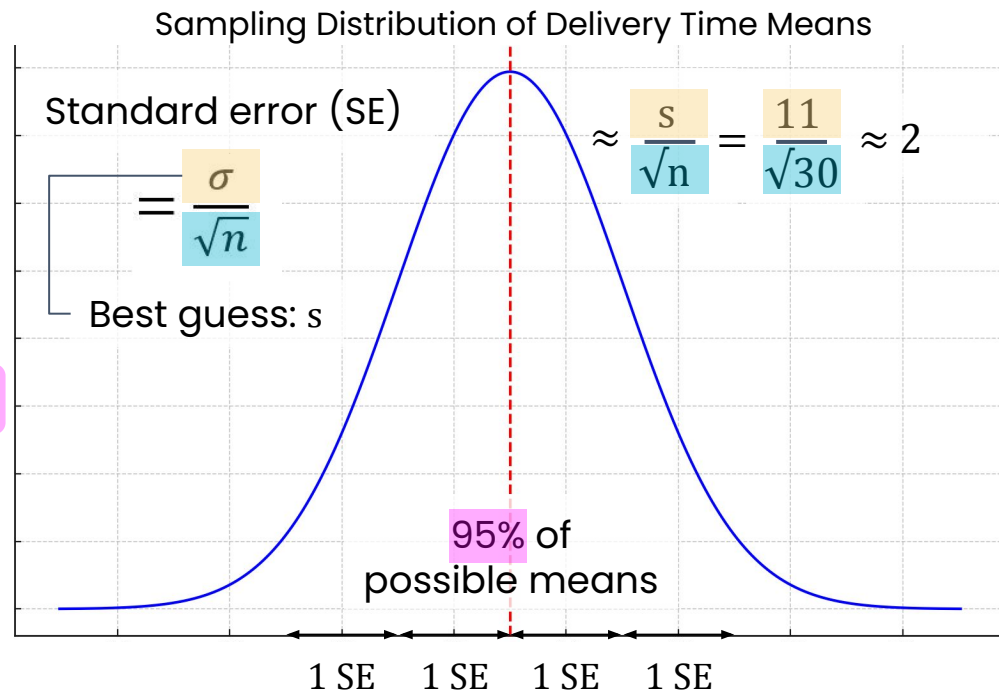
$$\begin{aligned}\bar{x} - 2 \times \text{SE} &= 43 - 2 \times 2 = 39 \\ \bar{x} + 2 \times \text{SE} &= 43 + 2 \times 2 = 47\end{aligned}$$

Variability

Sample size

Confidence

With 95% confidence, the true mean delivery time is between 39 and 47 minutes.



Confidence interval



- Range of values used to estimate a population parameter
- Quantifies the uncertainty of an estimate by the relative width of the range

Broader range
High uncertainty

Narrow range
More precision

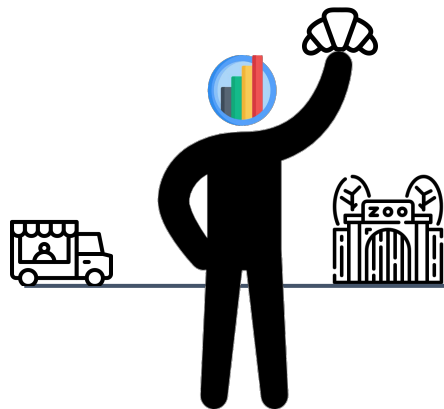


Understanding the possible average delivery time can help with precise scheduling

Scenario

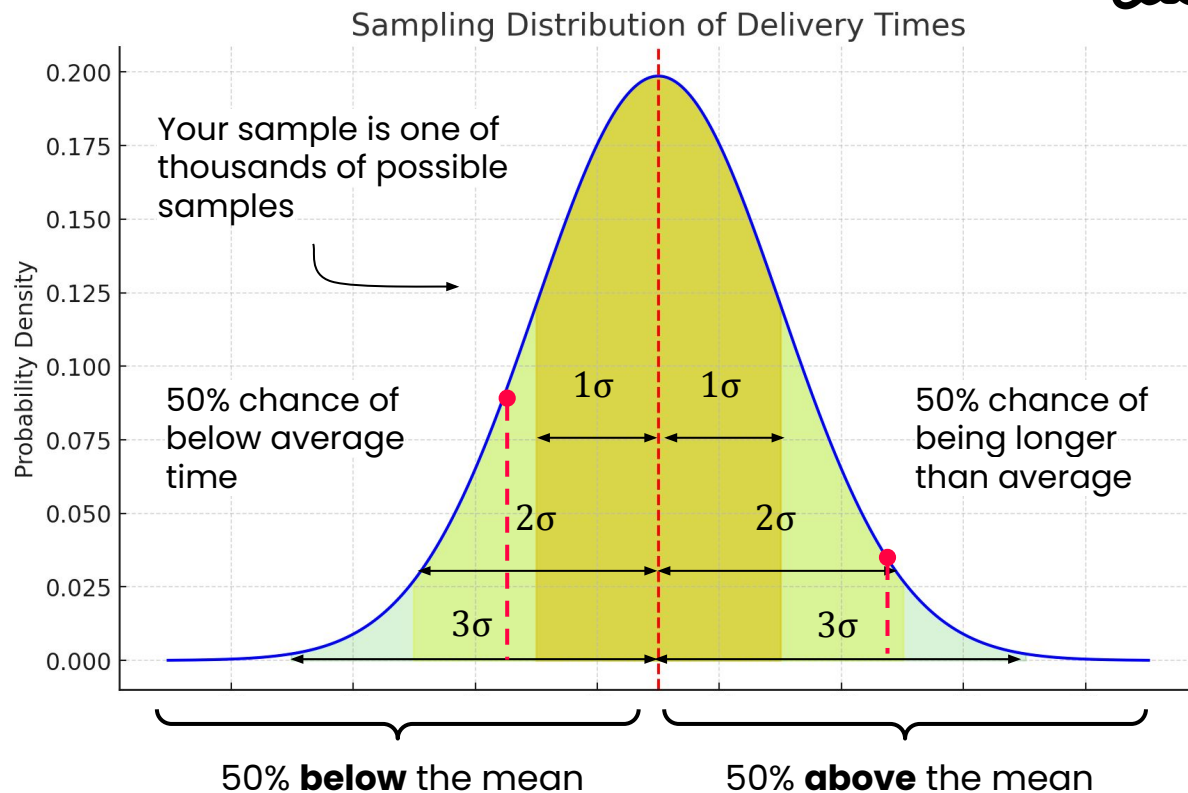


Before 7am



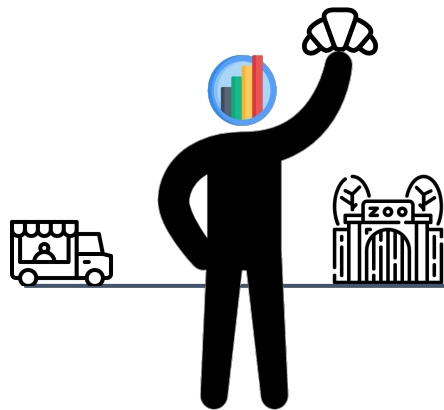
You

Data Analyst



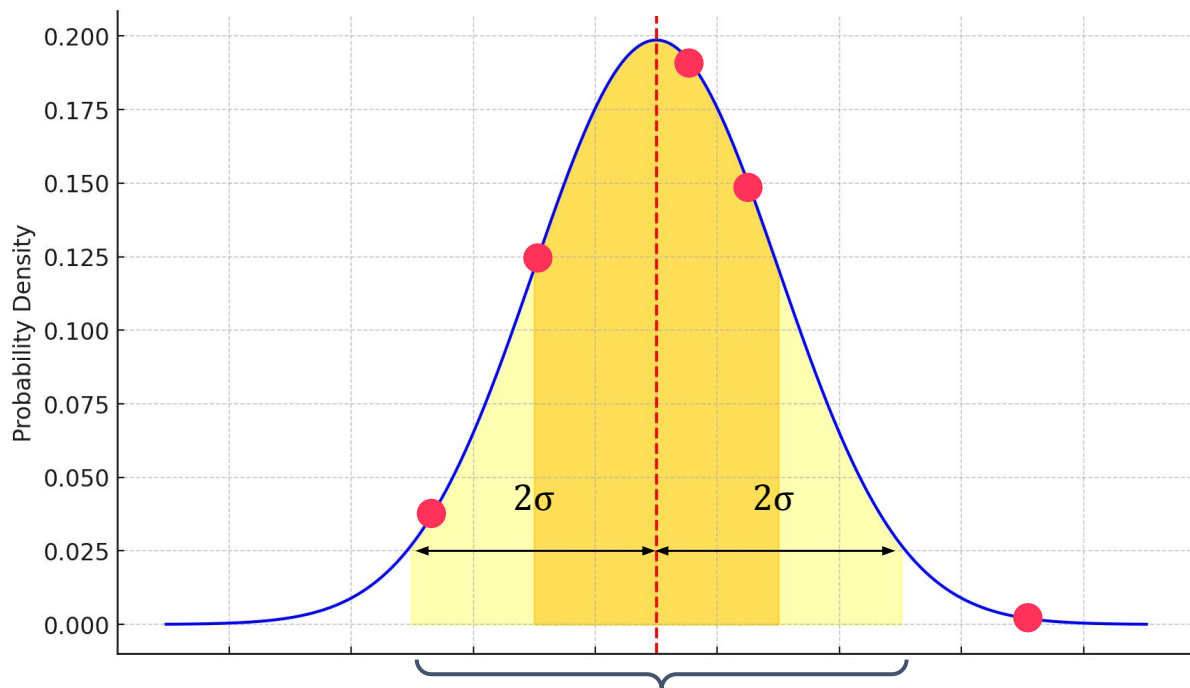
Scenario

 Before 7am



You
Data Analyst

Probability that sample mean 43 is
within **2 standard deviations** of μ ?



Based on the two sigma rule: **95% chance**

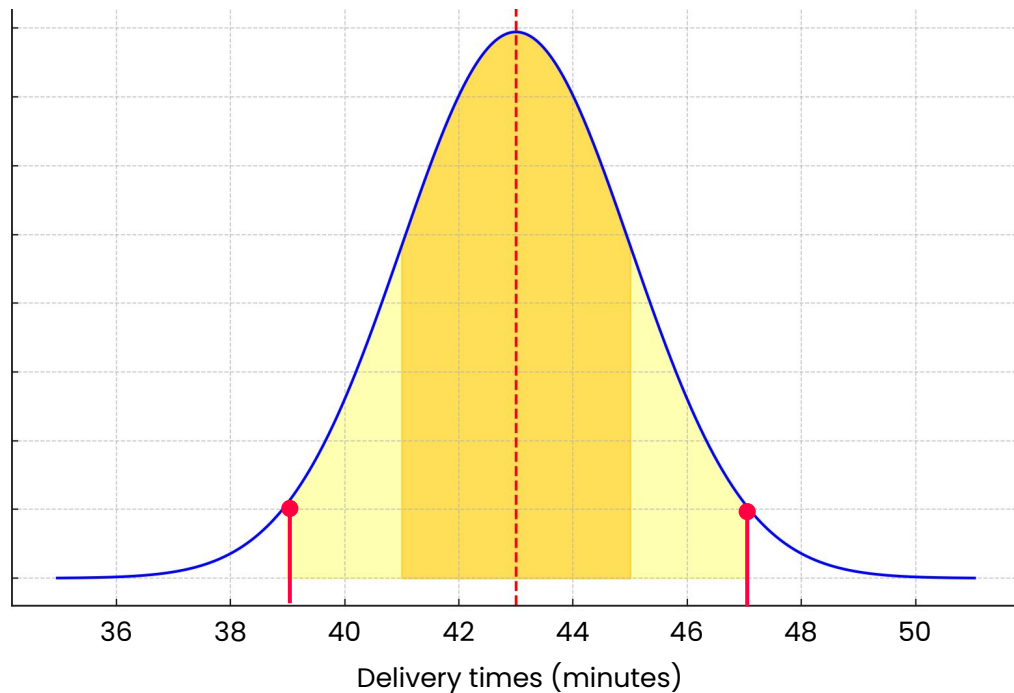
Confidence intervals

Mechanisms of confidence
intervals

Example

- 95% confidence level reflects how reliable your estimation method is
- A method that's designed to be right about 95% of the time
- Population parameter is:
 - Fixed, but unknown to you
 - In this particular interval or it's not

With 95% confidence, the true average delivery time is between 39 and 47 minutes



Confidence interval for the population mean

$$\bar{x} \pm z \times \left(\frac{s}{\sqrt{n}} \right)$$

Where:

- $\bar{x}$ - sample mean
- s - sample standard deviation
- n - sample size
- z - z-score value from the standard normal distribution

z-score

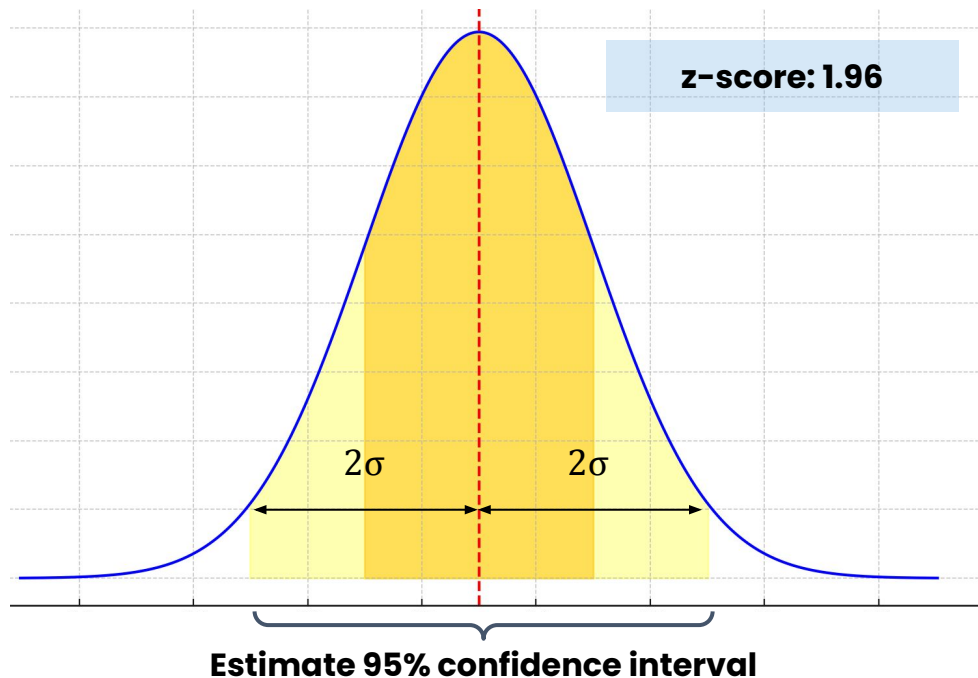
- Controls how “confident” you are that interval contains the population mean
- Number of standard deviations from mean in standard normal distribution

Confidence interval for the population mean

To calculate a **95% confidence interval**:

$$\bar{x} \pm 1.96 \times \left(\frac{s}{\sqrt{n}} \right)$$

- The two sigma rule is just an estimate
- $z = 2$ is associated with slightly higher confidence than 95%
- Use $z = 1.96$ to be more precise



Confidence interval for the population mean

To calculate a **95% confidence interval**:

$$\bar{x} \pm 1.96 \times \left(\frac{s}{\sqrt{n}} \right)$$

Margin of error

- Constructs the confidence interval
- Helps gauge the precision of the estimate

Confidence interval	z score
90%	1.645
95%	1.96
99%	2.576



- ↑ Higher level of confidence
- ↑ Use a higher z score
- ↑ Generate a wider confidence interval

Choosing a confidence level

90% confidence level

Used when missing true value is less important

Estimate avg. user rating



Users on average rate a new feature **7.2 out of 10**

- ✓ A less precise estimate is acceptable to help make initial decisions

95% confidence level

Balances confidence with potential for error

Estimate avg. delivery time



The average delivery time is between **39 and 47 minutes**

- ✓ Don't need to schedule too much buffer time

99% confidence level

Used when you want to minimize the risk of error

Estimate pollution in a river



Estimating pollution in a river

- ✓ Can help reduce risk of harmful impact

Confidence intervals

Understanding
margin of error

Margin of error

$$z \times \left(\frac{s}{\sqrt{n}} \right)$$

Determines how wide your confidence interval is

✓ A **narrower confidence interval** is more desirable because it means you have a **more precise** estimate

Depends on three factors:

- 1 Desired confidence level
- 2 Standard deviation
- 3 Sample size

i.e. the amount of variability in your data



Desired confidence level

90% confidence level — $z = 1.645$

$$z \times 1 = 1.645 \times 1 = 1.645$$

95% confidence level — $z = 1.96$

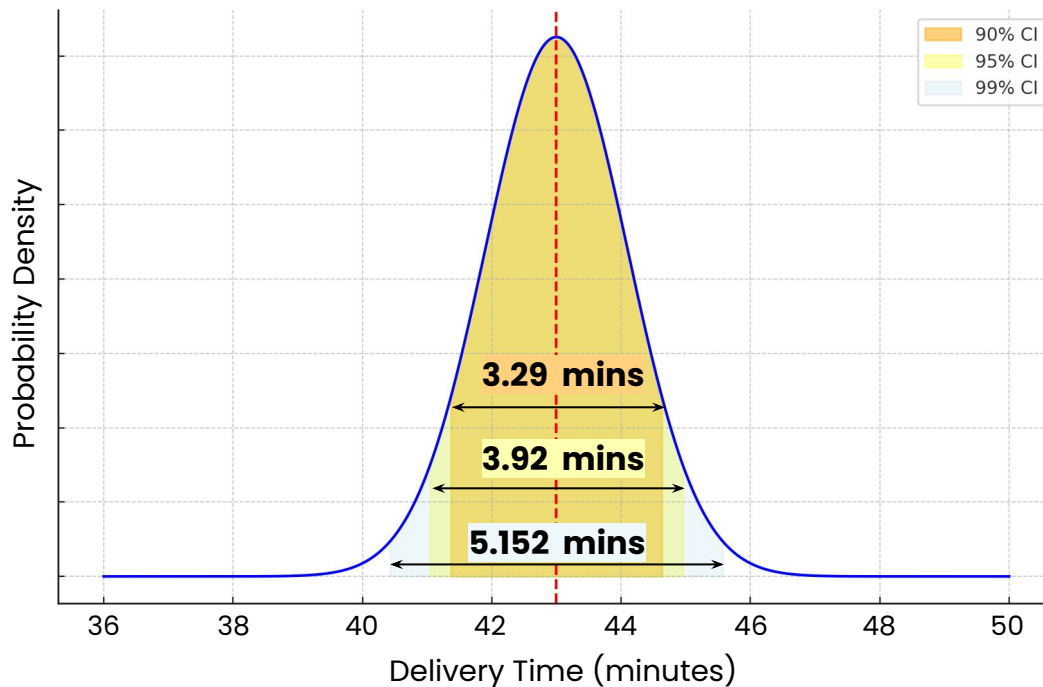
$$z \times 1 = 1.96 \times 1 = 1.96$$

99% confidence level — $z = 2.576$

$$z \times 1 = 2.576 \times 1 = 2.576$$

$$z \times \left(\frac{s}{\sqrt{n}} \right) = z \times \left(\frac{10}{\sqrt{100}} \right) = z \times \left(\frac{10}{10} \right) = z \times 1$$

Confidence Intervals for Delivery Times



Width vs. Confidence level $z \times \left(\frac{s}{\sqrt{n}}\right) = z \times \left(\frac{10}{\sqrt{100}}\right) = z \times \left(\frac{10}{10}\right) = z \times 1$

90% confidence level

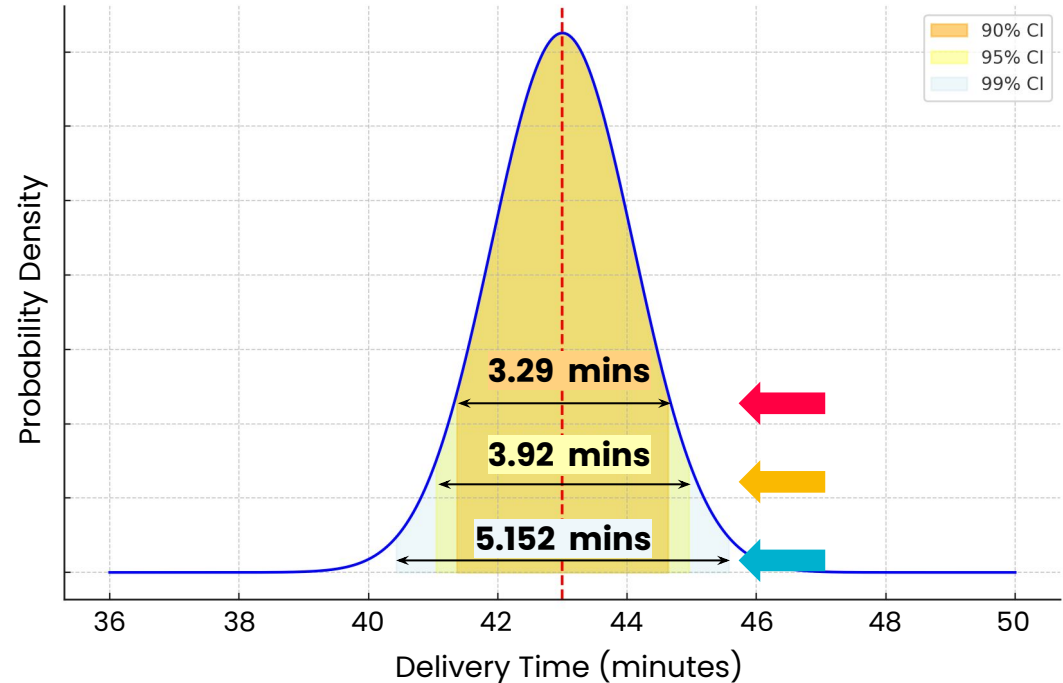
95% confidence level

- 19% wider
- Gain 5% in relative confidence

99% confidence level

- 31% wider
- Gain 4% in relative confidence

Confidence Intervals for Delivery Times



Variability

s = 10 

$$1.96 \times \left(\frac{10}{\sqrt{100}} \right) = 1.96 \times \left(\frac{10}{10} \right) = 1.96 \times 1 = 1.96$$

s = 20

$$1.96 \times \left(\frac{20}{\sqrt{100}} \right) = 1.96 \times \left(\frac{20}{10} \right) = 1.96 \times 2 = 3.92$$

s = 30 

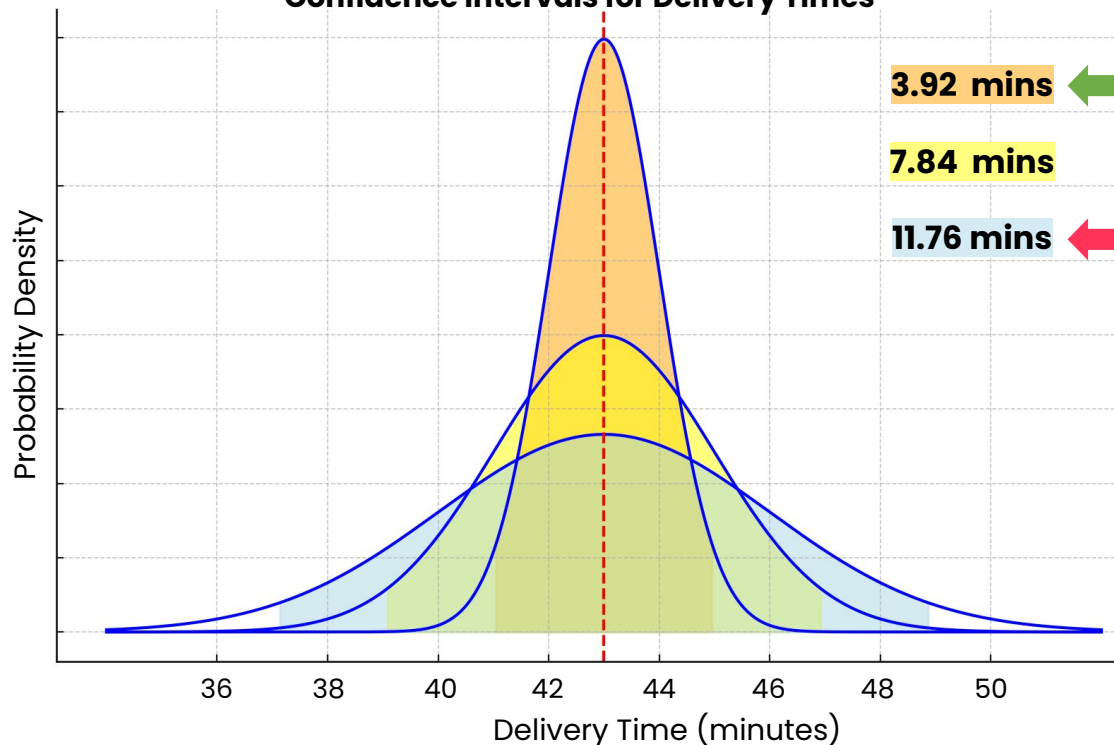
$$1.96 \times \left(\frac{30}{\sqrt{100}} \right) = 1.96 \times \left(\frac{30}{10} \right) = 1.96 \times 3 = 5.88$$

If data has a lot of variability, expect a less precise estimate.

95% confidence interval

$$z \times \left(\frac{s}{\sqrt{n}} \right) = z \times \left(\frac{s}{\sqrt{100}} \right) = 1.96 \times \left(\frac{s}{\sqrt{100}} \right)$$

Confidence Intervals for Delivery Times



Variability



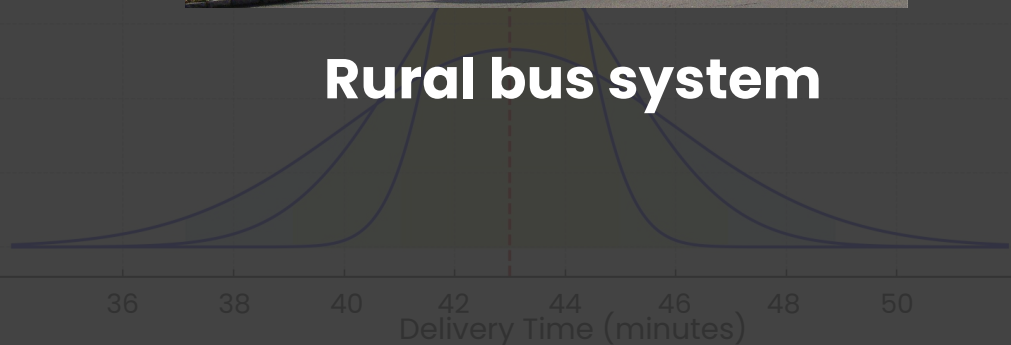
City bus system



Smaller variability makes it easier to estimate the true average arrival time



Rural bus system



Sample size

$$z \times \left(\frac{s}{\sqrt{n}} \right) \leftarrow 1.96 \times \left(\frac{10}{\sqrt{n}} \right)$$

n = 100

$$1.96 \times \left(\frac{10}{\sqrt{100}} \right) = 1.96 \times \left(\frac{10}{10} \right) = 1.96 \times 1 = 1.96$$

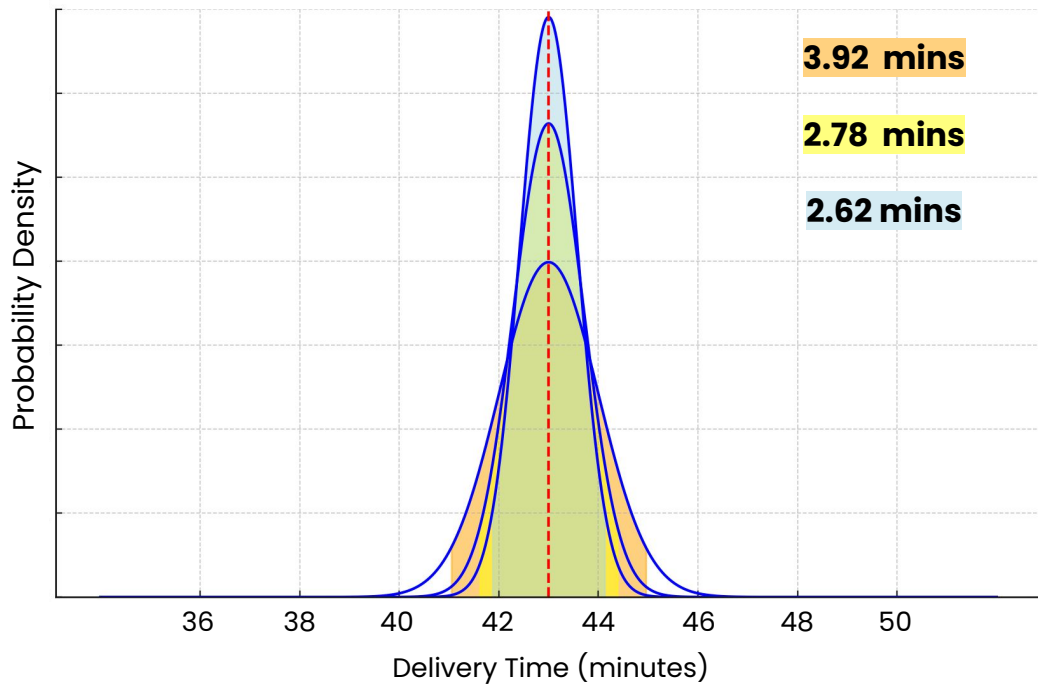
n = 200

$$1.96 \times \left(\frac{10}{\sqrt{200}} \right) = 1.96 \times \left(\frac{10}{14.1} \right) = 1.96 \times 0.71 \approx 1.39$$

n = 300

$$1.96 \times \left(\frac{10}{\sqrt{300}} \right) = 1.96 \times \left(\frac{10}{17.3} \right) = 1.96 \times 0.58 \approx 1.13$$

Confidence Intervals for Delivery Times



Sample size

$$z \times \left(\frac{s}{\sqrt{n}} \right) \leftarrow$$

A larger sample:

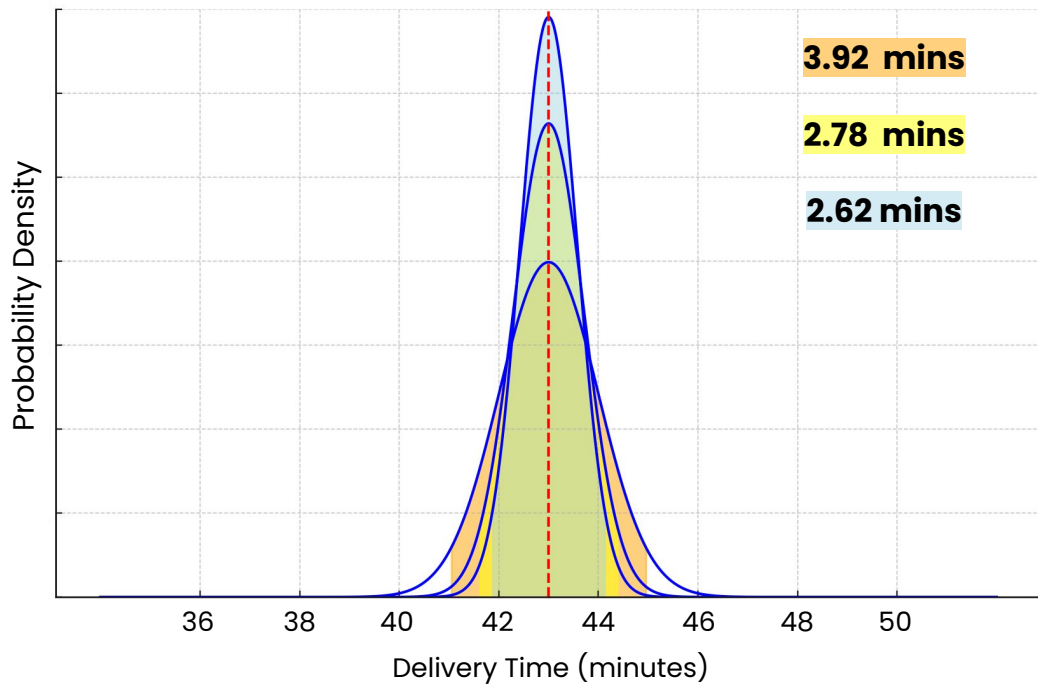
- Allows you to construct a narrower confidence interval
- Reduces the size of the margin of error
- Produces diminishing returns

n = 100

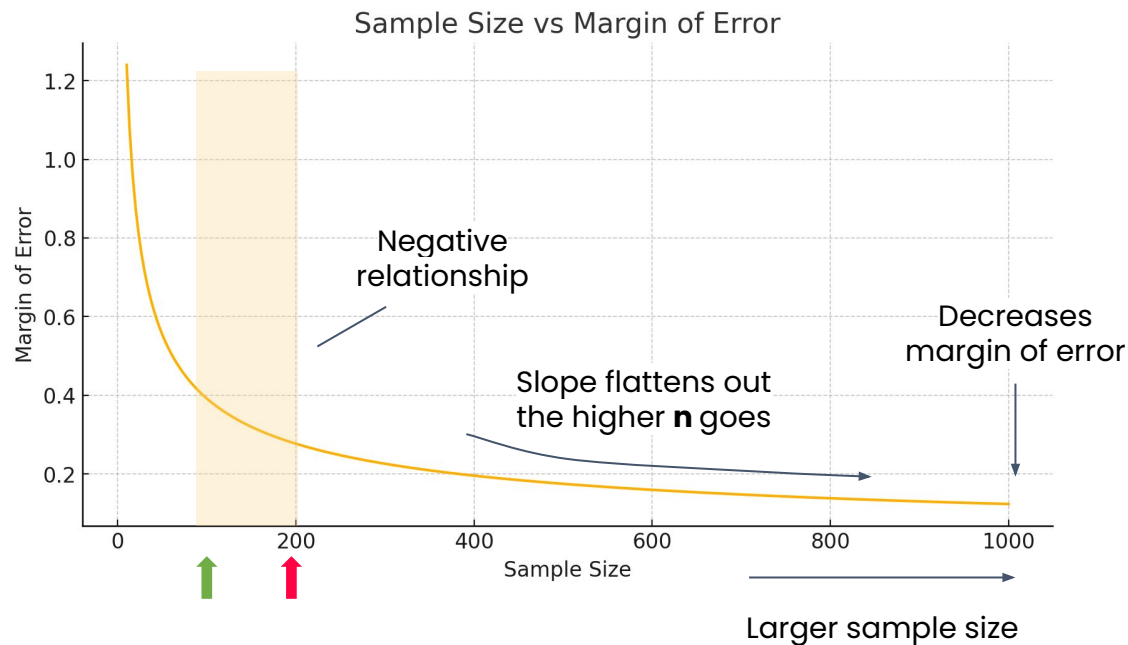
n = 200 → Reduced interval by 29%

n = 300 → Reduced interval by 6%

Confidence Intervals for Delivery Times



Sample size



- This relationship is a basic fact of statistics
- Often need a relatively small sample for very large population 
- Adding more and more data will narrow interval
- In very high sample sizes, you will observe diminishing impact

Summary

You can achieve a narrower confidence interval in three ways:

- 1 By lowering your confidence level, which can increase your chances of missing the true value
- 2 By working with data that has less variability, which is generally out of your control
- 3 By increasing your sample size, though this approach has diminishing returns

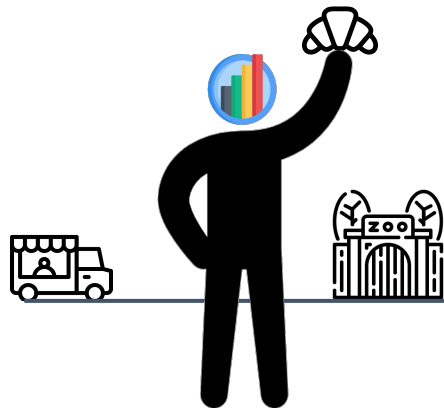
Confidence intervals

Demo: confidence intervals
for means

Confidence intervals

Confidence intervals
for proportions

Scenario



You
Data Analyst



Problem: “What proportion of deliveries are on time?”



Deliveries that made
it to the zoo by 7am



p

True proportion of
on-time deliveries



Collect a sample of 30 deliveries



Record if they were:



On time



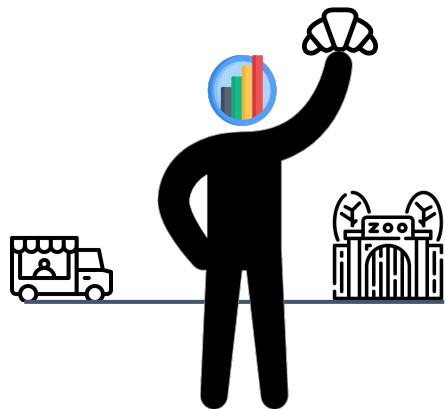
Not on time

$\hat{p}$



Estimate for the true
proportion of p

Scenario



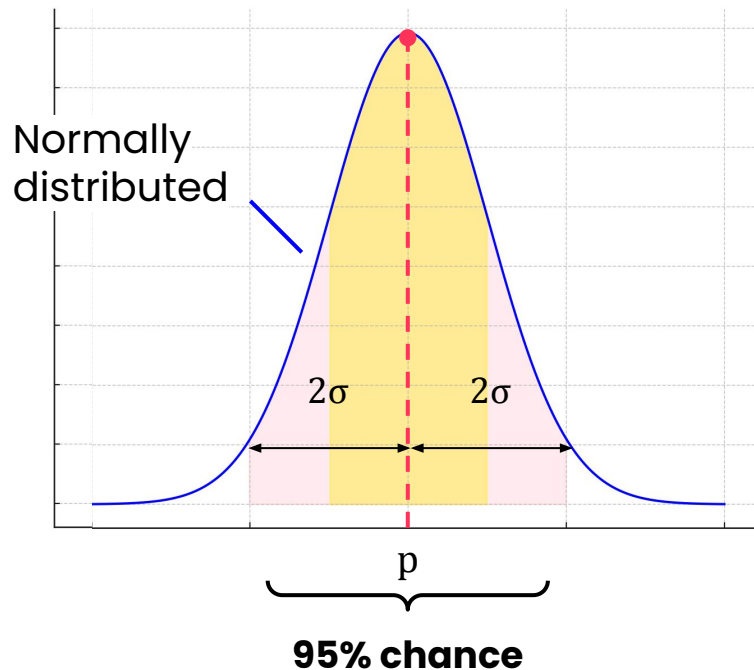
You
Data Analyst

$$\hat{p} = 0.6 \rightarrow$$



18 out of 30
on-time deliveries

Sampling distribution of $\hat{p}$



Standard error

The standard error for the proportion is:

$$\sqrt{\frac{p(1-p)}{n}}$$

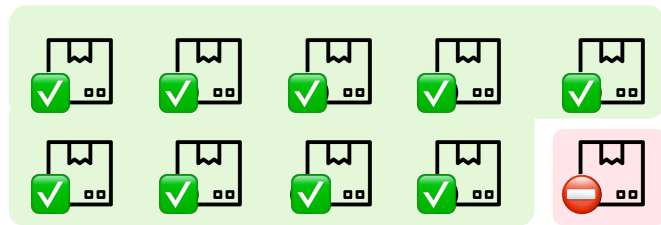
Square root

Use $\hat{p}$ to estimate

Sample size

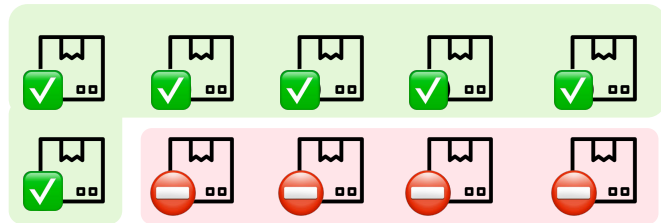
- $p(1-p)$ represents the variability
- Higher variability when $\hat{p}$ is closer to 0.5
- Take the square root so number is at the original scale

Low variability



$$\hat{p} = 0.9 \longrightarrow \hat{p}(1 - \hat{p}) = 0.09$$

Higher variability



$$\hat{p} = 0.6 \longrightarrow \hat{p}(1 - \hat{p}) = 0.24$$

More even mix of success and failures

Confidence interval

✓ $\hat{p}$ - estimate of true proportion of p

✓ $\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$ - standard error

Confidence interval for the mean:

$$\bar{x} \pm z \times \frac{s}{\sqrt{n}}$$

The interval is defined as:

Margin of error

$$\hat{p} \pm z \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

represents the uncertainty
in your estimate

Confidence interval

✓ $\hat{p}$ - estimate of true proportion of p

✓ $\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$ - standard error

Confidence interval for the mean:

$$\bar{x} \pm z \times \frac{s}{\sqrt{n}}$$

The interval is defined as:

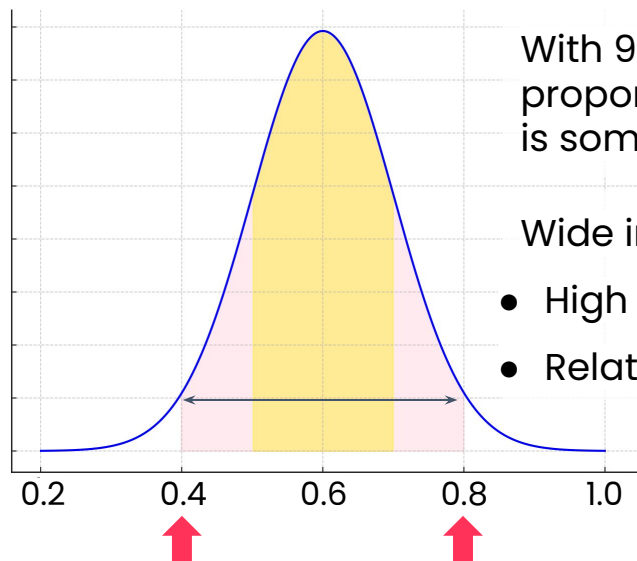
$$\hat{p} \pm z \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Confidence interval

$$\hat{p} = 0.6$$



18 out of 30
on-time deliveries



With 95% confidence, the true proportion of on time deliveries is somewhere within this range

Wide interval is reflected in:

- High variability of the data
- Relatively small sample size

$$= \hat{p} \pm z \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$= 0.6 \pm z \times \sqrt{\frac{0.6(1 - 0.6)}{n}}$$

$$= 0.6 \pm z \times \sqrt{\frac{0.6 \times 0.4}{n}}$$

$$= 0.6 \pm 1.96 \times \sqrt{\frac{0.6 \times 0.4}{n}}$$

$$= 0.6 \pm 1.96 \times \sqrt{\frac{0.6 \times 0.4}{30}}$$

$$= (0.4247, 0.7753)$$

Confidence intervals

Demo: confidence
intervals for proportions

Confidence intervals

Interpretation with LLMs

Confidence intervals

Simulating random
sampling with LLMs

Confidence intervals

Inference and visualization
with LLMs